

A Length-instruction Following

We analyze to what extent length-controlled summarization is a consistent probe for content selection. Ideally, we expect the following behavior of summarization models: (1) the generated summary length matches approximately the target length, and (2) as we increase the length budget, summaries should provide all content of the shorter version in addition to expanding on it. We define two measures for these desiderata.

Target length ratio (TLR). We quantify the length deviation of a generated summary (s_l) from the target word count (l) as follows:

$$\text{TLR}(s_l) = \frac{|s_{d,l}|}{l}. \quad (4)$$

Where $|\cdot|$ is the summary length (whitespace separated tokens). A value of 1 indicates perfect length match, while values greater or smaller than 1 indicate over- or under-generation, respectively.

Incremental consistency (IC). Longer summaries should contain a proper superset of claims found in the adjacent shorter version. Formally, for each document d and topic t recall that we have a set of atomic claims A_t (§ 2.2). We first identify the set of claims that are entailed at least once across any summary length:

$$A_{\text{entailed}}(d, t) = \{a \in A_t \mid \exists l \in L, e(a, s_{d,l}) = 1\},$$

where e is an NLI model indicating whether claim a is entailed by summary $s_{d,l}$ of length l . Next, we determine if a claim is included consistently across increasing summary lengths (monotonicity condition).

$$e(a, s_{d,l_1}) \leq e(a, s_{d,l_2}) \quad \forall l_1 < l_2$$

We then define the set of consistent claims where this condition holds:

$$A_{\text{consistent}}(d, t) = \{a \in A_{\text{entailed}}(d, t) \mid \text{monotonicity holds } \forall l \in L\}.$$

Finally, the overall incremental consistency for summaries of a corpus D is given as the fraction of consistent claims:

$$\text{IC}(D) = \frac{\sum_{d \in D} \sum_{t \in T} |A_{\text{consistent}}(d, t)|}{\sum_{d \in D} \sum_{t \in T} |A_{\text{entailed}}(d, t)|}. \quad (5)$$

This metric ranges from 0 to 1, where 1 indicates perfect monotonicity (longer summaries always include all information found in shorter ones).

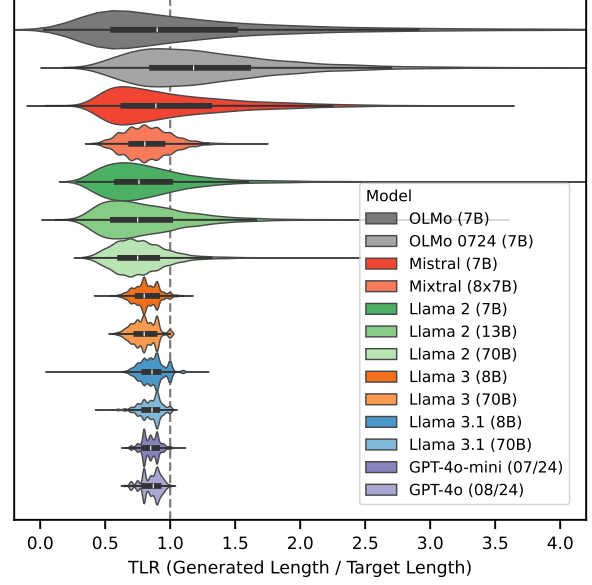


Figure 5: Distribution of target length ratios over all generated summaries (aggregating lengths and datasets).

Do models meet the target length? We find that all models generally undershoot the length target (Figure 5). However, more recent models match the target length more closely and consistently, showing a clear scaling effect. The best performing models are Llama 3.1 and GPT-4o, while OLMo is unable to follow length-instructions, presumably because this was not part of the instruction tuning data. Surprisingly, we do not find substantial differences across datasets. This suggests that the ability of models to follow length-instructions is mostly invariant to the input document length, even if they are considerably long (e.g., meeting transcripts). See Figure 10 for an analysis of summary length stratified by dataset and target length.

How incrementally consistent are summaries?

We report the average incremental consistency by dataset and model in Figure 6. We observe that all models are substantially more consistent than the random summarization baseline. Furthermore, incremental consistency decreases with more difficult datasets, likely because there is more freedom on what content to include in a summary. Similar to the ability of following length instructions, we observe a scaling effect where stronger models have a higher incremental consistency.

Influence of temperature sampling. The main results in this paper are obtained with a temperature of $\tau = 0.3$. To assess how temperature affects summary length and incremental consistency, we perform a temperature sweep on the RCT dataset

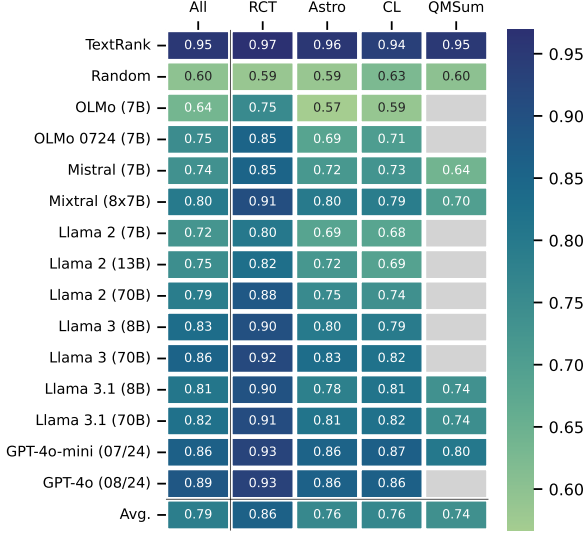


Figure 6: Incremental consistency by model and dataset.

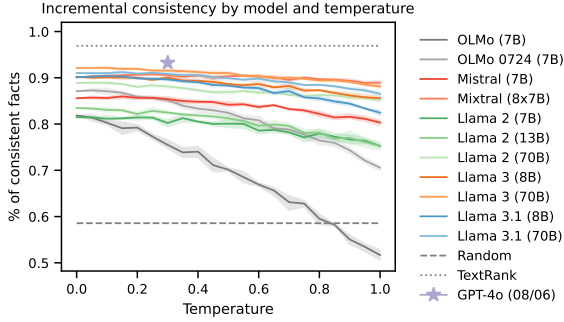


Figure 7: Incremental consistency by temperature.

for all open-weights models (20 settings in $[0, 1]$). Surprisingly, higher temperatures do not affect the *average* summary length on a dataset-level, but lead to greater variance at the document level (up to 10% length difference between generations, Figure 9). Furthermore, higher temperatures lead to a slight decline in incremental consistency for all models that adequately follow length instructions (a drop of 1% to 9%, Figure 7).

Summary. Overall, we find that strong models are able to follow length-instructions and that they consistently expand the summary content with increasing length budgets. As our salience analysis assumes this behavior of models, it may be less reliable for weaker models (OLMo, Mistral, Llama 2).

B Salience Analysis

The corpus-level salience analysis for PubMed, Astro, CL, and QMSum is given in Figure 11, Figure 12, Figure 13, and Figure 14, respectively. We also provide a fully-worked example of the content salience analysis in Figure 18.

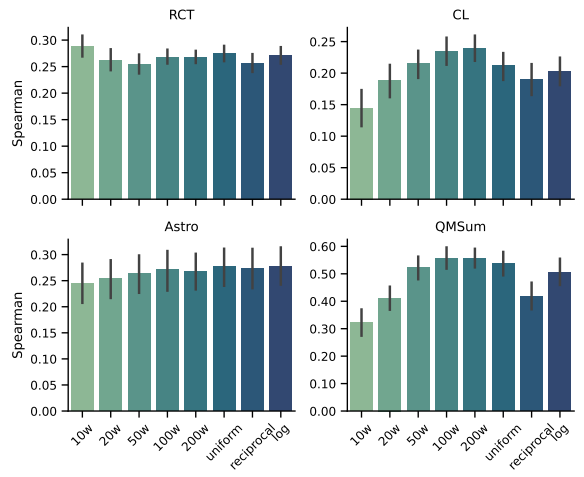


Figure 8: Correlation of different salience scores with human salience. Here we aggregate over all LLMs which showed similar trends.

C Ablation: Salience Score

We analyze how different salience scores derived from the CSM correlate with human salience. Recall that the $\text{CSM}(D)_{t,l}$ tracks the average answerability of question $t \in T$ at summary length $l \in L = \{10, 20, 50, 100, 200\}$. We take raw salience scores at each summary length. Additionally, we calculate several question-wise aggregate scores. Intuitively, questions which are more answerable at shorter summaries score higher under the aggregated scheme. Formally, we aggregate scores as follows:

$$\text{CSM}_{\text{agg}}(D)_t = \frac{\sum_{l \in L} w_l \cdot \text{CSM}(D)_{t,l}}{\sum_{l \in L} w_l},$$

where w_l is a weighting term. We experiment with three weighting functions: uniform ($w_l = 1$), reciprocal length ($w_l = 1/l$), and logarithmic decay ($w_l = 1/\log(1 + l)$).

Figure 8 shows the Spearman rank correlation coefficient (ρ) with human salience for each salience score. Overall, we find that all salience scores correlate similarly with human salience ratings on *RCT* and *Astro*, while the 200 words salience score correlates most strongly on *CL* and *QMSum*.

D Ablation: Effect of Answer Length on Question Salience

We calculate question salience as the fraction of answer claims entailed by the summary (Equation 3). Naturally, some questions can be answered succinctly (e.g., “What is the goal of the study?”) while others require more elaboration (e.g., “What

Summary	RCT	Astro	CL	QMsum
10 words	-0.03	-0.11**	-0.07**	-0.22**
20 words	-0.10**	-0.17**	-0.10**	-0.31**
50 words	-0.18**	-0.21**	-0.19**	-0.36**
100 words	-0.26**	-0.28**	-0.22**	-0.41**
200 words	-0.31**	-0.31**	-0.26**	-0.45**

Table 5: Spearman rank correlation between *answer length* and *question salience* for summaries generated by Llama 3.1 (70B). Significance: ** ($p < 0.01$).

were the detailed findings?”). This raises the question how answer length influences salience. To better understand this relationship, we compute the Spearman rank correlation between answer length (measured in whitespace-separated tokens) and question salience. Table 5 reports results for summaries generated by Llama 3.1 (70B), with similar trends for other models.

We observe a weak negative correlation between answer length and question salience for shorter summaries (10–20 words), and a weak-to-moderate negative correlation for longer ones (≥ 100 words). This suggests that answer length explains some variance in question salience, but cannot fully account for it. A hypothesis is that model-generated summaries are abstractive, possibly conveying information more densely than the reference answers.

E Pilot Study: Human Salience in News Summaries

To test the generality of our framework, we use it as a tool to analyze the salience notions encoded in human-written summaries from a standard summarization benchmark. We focus on news articles from the CNN/DM dataset (Hermann et al., 2015).

Method. First, we run question generation over a sample of 200 random documents to identify QUDs for this domain (see Steps 1 and 2 in § 2.2). Next, we use the resulting questions to analyze human summaries (see Steps 3 and 4 in § 2.2). Since each article in CNN/DM has only one reference summary, we select (document, summary) pairs with summaries approximating our target lengths of $L = \{10, 20, 50, 100, 200\}$ words, allowing for a delta of $\pm 10\%$.⁹ Finally, from each length bucket, we draw a random sample of 200 (document, summary) pairs for analysis.

Results. We present the content salience map for both human and model summaries in Figure 15. We observe consistent trends in question salience.

⁹Of the 287,113 documents in the CNN/DM training set, 0.28/1.60/26.84/3.50/0.04% fall into the respective buckets.

For example, questions about the main event ($Q1$) or its magnitude ($Q13$) and consequences ($Q6$) consistently achieve higher scores than more detailed questions about reactions ($Q7$), expert opinions ($Q10$) or additional stakeholders ($Q3$). While the salience scores of human and model summaries are not directly comparable due to differing document samples, they exhibit similar trends.

In sum, this pilot study demonstrates the versatility of the framework, and suggests that it could be used in future work to understand human notions of salience on a larger scale.

F Responsible NLP Considerations

Compute Requirements. Experiments were conducted on NVIDIA A100 80GB GPUs, requiring approximately 20 GPU hours per dataset, and an additional 360 GPU hours for the temperature sweep on the RCT dataset, totaling 440 GPU hours. We ran inference using VLLM (docs.vllm.ai). GPT-4o models were accessed through the OpenAI API with inference costs $\leq 100\$$.

Salience Annotation Study. Participants joined on a volunteer basis, gave informed consent and agreed that their annotations will be shared in anonymized form in the paper repository. According to our institutional policies, this study did not require institutional review board (IRB) approval.

Data Licensing. We obtain RCT abstracts in accordance with fair use principles through the PubMed Entrez API.¹⁰ Related work sections of CL and Astro papers were collected via the arXiv API.¹¹ While the majority of papers on arXiv is published under the arXiv license¹² retaining *copyright* with the original author(s), the *use* of paper contents for research is explicitly granted and encouraged in the arXiv API terms & conditions.¹³ We reused meeting transcripts from QM-Sum (Zhong et al., 2021).¹⁴ All meeting transcripts are under an open use license, such as CC BY 4.0 (academic meetings and product meetings) or Open Government License Version 3 (parliament committee meetings).^{15,16,17,18}

All URLs accessed 2025-05-15.

¹⁰www.ncbi.nlm.nih.gov/home/develop/api/

¹¹info.arxiv.org/help/api/index.html

¹²arxiv.org/licenses/nonexclusive-distrib/1.0/license.html

¹³info.arxiv.org/help/api/tou.html

¹⁴github.com/Yale-LILY/QMSum

¹⁵creativecommons.org/licenses/by/4.0/legalcode

¹⁶groups.inf.ed.ac.uk/ami/icsi/license.shtml

¹⁷groups.inf.ed.ac.uk/ami/corpus/license.shtml

¹⁸www.nationalarchives.gov.uk/doc/open-government-licence/